

MITRE ATT&CK; Mapping Appendix

Lab 1 — Windows Authentication Monitoring

Technique: T1078 — Valid Accounts

Tactic: Persistence / Initial Access

Simulated failed and successful login attempts to validate authentication logging.

Lab 2 — Malware Detection Using EICAR

Technique: T1204 — User Execution

Tactic: Execution

Simulated malware execution using the EICAR test file.

Lab 3 — Brute Force Attack Simulation

Technique: T1110 — Brute Force

Tactic: Credential Access

Performed repeated authentication attempts using Hydra.

Lab 4 — PowerShell Activity Monitoring

Technique: T1059 — Command and Scripting Interpreter (PowerShell)

Tactic: Execution

Executed PowerShell commands to generate monitored logs.

Lab 5 — File Integrity Monitoring

Technique: T1565 — Data Manipulation

Tactic: Impact

Created and modified files to simulate integrity tampering.